PROJECT PRESENTATION & SYSTEM MANUAL

Integrated E-Commerce Cafe System & 24 Seasonal Botanical Blends

Engineering Feasibility, Software Architecture, Server-Side Dynamic QRIS Settlement Governance, and Barista Operational Precision at CHANGKEER ATELIER.

CHANGKEER BOARD OF FOUNDERS (THE 5 CO-FOUNDERS)

No	Founder Name	Role	Executive Operational Focus
01	**ALMAIDAH CHAERUDIN PUTRI**	Co-founder	Championing honest daily hydration without synthetic shortcuts.
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SECTION 01 • BUSINESS CONTEXT & BOTANICAL VISION

Industry Landscape & The 24 Solar Terms Philosophy

Pioneering a wholesome botanical hydration alternative synchronized with human circadian biological rhythms.

Challenges in Modern Beverage Retail

- **Excessive Synthetic Sweeteners:** Commercial beverages are saturated with artificial flavoring agents, high-fructose corn syrups, and synthetic non-dairy creamers harmful to long-term health.
- **Stimulant Caffeine Dependency:** Excessive coffee intake often triggers gastric acidity, nervous palpitations, and mid-afternoon energy crashes.
- **Absence of Mindful Spaces:** Standard cafes operate as loud, fast-turnover transaction hubs that fail to provide restorative environments for contemplation.

CHANGKEER 24 Solar Terms Solution

- **100% Pure Botanical Tisanes:** Crafted exclusively from whole organic flowers, wild her and rimpang roots; completely caffeine-free, gentle on digestion, and soothing.
- **Harmonized with 24 Solar Terms:** 24 recipes balanced with natural cycles (Spring for vitality renewal, Summer for heat dispersal, Autumn for nervous grounding, Winter for immune support).
- **Calibrated Natural Sweetening:** Standardized dispensations of raw forest honey or sorghum nectar measured strictly via milliliter jiggers.

Sourcing Integrity: Zero synthetic flavoring essences; all raw botanicals sourced directly from cooperative farmers in Wonosobo, Garut, and Puncak.

SECTION 02 • INTEGRATED RETAIL ARCHITECTURE

Service Architecture: Guest Self-Service & Bar Efficiency

Unifying app-free mobile web ordering with real-time Kitchen Display System (KDS) execution.

Autonomous Guest Experience (Mobile Web)

- **Zero-Install Access:** Guests scan table QR codes; the high-performance web app renders instantly in mobile browsers without downloading native apps.
- **Precision Customization:** Guests personalize temperature (hot ceramic pot or craft iced shaken), natural sweetness levels (0% to 100%), and organic dried flower & fruit infusions.
- **Dynamic Exact-Amount QRIS:** Payment codes pre-populate the exact bill total into mobile banking apps, eliminating manual input mistakes.

Bar Station Operations (Kitchen Display System)

- **Dedicated Brewing Craft:** Baristas are freed entirely from writing manual paper chits or handling physical cash transactions.
- **Strict FIFO Queue Management:** Tickets arrive in strict queue sequence with automate digital countdown timers and precise temperature guides.
- **Instant Status Sync:** Brewing progression updates instantly across both kitchen monito and customer smartphone screens.

Operational Impact: Eliminates counter ordering queues by over 70% while drastically improving extraction consistency across all seasonal blends.

SECTION 03 • SEASONAL FORMULATIONS (1/4)

Spring Season Catalog: Vitality Renewal & Metabolic Cleansing

Solar Terms 01 to 06 — Light floral formulations anchored by jasmine, rose buds, and tender shoots for natural cellular awakening.

No	Solar Term	Blend Formulation	Key Ingredients	Brew Standard	Price	Flavor Profile & Notes
01	 **Start of Spring** 立春	**First Bloom**	jasmine flowers, dried pear, goji	82°C • 3.5 Min	**Hot 24K** **Iced 26K**	Light, floral and a little fruity.
02	 **Rain Water** 雨水	**Rain Garden**	rose buds, ginger, red dates	98°C • 6.0 Min	**Hot 20K** **Iced 22K**	Rose up front, soft ginger warmth.
03	 **Awakening of Insects** 惊蛰	**Green Wake**	mint, lemongrass, pandan	88°C • 4.0 Min	**Hot 15K** **Iced 17K**	Fresh and grassy, with a pandan finish.
04	 **Spring Equinox** 春分	**Even Day**	chrysanthemum, rose buds, goji	92°C • 5.0 Min	**Hot 22K** **Iced 24K**	Half floral, half fruity.
05	 **Clear and Bright** 清明	**Clear Sky**	butterfly pea, lemongrass, lemon	88°C • 4.0 Min	**Hot 18K** **Iced 20K**	Blue in the cup, purple once the lemon goes in.
06	 **Grain Rain** 谷雨	**Barley Field**	roasted barley, osmanthus	88°C • 4.0 Min	**Hot 20K** **Iced 22K**	Toasty and nutty, with a honeyed flower note.

Serving options: Ceramic hot infusion pot (standard) or craft iced shaken tea (+IDR 2,000). Zero synthetic essences; 100% caffeine-free.

SECTION 03 • SEASONAL FORMULATIONS (2/4)

Summer Season Catalog: Active Hydration & Biological Cooling

Solar Terms 07 to 12 — Refreshing mountain mint, butterfly pea, and native citrus designed for thirst quenching and heat dispersal.

No	Solar Term	Blend Formulation	Key Ingredients	Brew Standard	Price	Flavor Profile & Notes
07	 **Start of Summer** 立夏	**Hibiscus Sun**	hibiscus, orange peel, mint	92°C • 5.0 Min	**Hot 18K** **Iced 20K**	Tart, ruby-red and very refreshing.
08	 **Grain Buds** 小满	**Pandan Lime**	pandan, lemongrass, lime leaf	88°C • 4.0 Min	**Hot 15K** **Iced 17K**	Green and citrusy.
09	 **Grain in Ear** 芒种	**Mango Jasmine**	dried mango, jasmine flowers	82°C • 3.5 Min	**Hot 26K** **Iced 28K**	Ripe fruit with a jasmine lift.
10	 **Summer Solstice** 夏至	**Longest Day**	chrysanthemum, lemon, mint	88°C • 4.0 Min	**Hot 18K** **Iced 20K**	Bright, clean and cooling.
11	 **Minor Heat** 小暑	**Cool Leaf**	mint, lime leaf, lemongrass	88°C • 4.0 Min	**Hot 15K** **Iced 17K**	Minty and sharp.
12	 **Major Heat** 大暑	**Chrysanthemum Lemon**	chrysanthemum, lemon, goji	88°C • 4.0 Min	**Hot 20K** **Iced 22K**	Our hot-season classic: mellow flowers, fresh citrus.

Serving options: Ceramic hot infusion pot (standard) or craft iced shaken tea (+IDR 2,000). Zero synthetic essences; 100% caffeine-free.

SECTION 03 • SEASONAL FORMULATIONS (3/4)

Autumn Season Catalog: Restorative Calm & Nervous Grounding

Solar Terms 13 to 18 — Whole chamomile blossoms, tropical cinnamon bark, and dried pear to ease mental stress and support deep sleep.

No	Solar Term	Blend Formulation	Key Ingredients	Brew Standard	Price	Flavor Profile & Notes
13	 **Start of Autumn** 立秋	**Osmanthus Pear**	osmanthus, dried pear, goji	88°C • 4.0 Min	**Hot 26K** **Iced 28K**	Apricot-scented flowers with soft pear.
14	 **End of Heat** 处暑	**Late Rose**	rose buds, hibiscus, dried apple	92°C • 5.0 Min	**Hot 22K** **Iced 24K**	Rosy and tart, like fruit jam.
15	 **White Dew** 白露	**White Dew**	white chrysanthemum, dried pear	88°C • 4.0 Min	**Hot 22K** **Iced 24K**	Delicate and smooth. Try it with honey.
16	 **Autumn Equinox** 秋分	**Golden Hour**	osmanthus, roasted barley, orange peel	88°C • 4.0 Min	**Hot 24K** **Iced 26K**	Toasty and golden.
17	 **Cold Dew** 寒露	**First Rain**	ginger, goji, red dates	98°C • 6.0 Min	**Hot 20K** **Iced 22K**	Warming ginger and naturally sweet dates, for the first wet evenings.
18	**Frost** 霜降	**Ginger Rose**	ginger, rose buds, cinnamon	98°C • 6.0 Min	**Hot 20K** **Iced 22K**	Spiced and floral.

Serving options: Ceramic hot infusion pot (standard) or craft iced shaken tea (+IDR 2,000). Zero synthetic essences; 100% caffeine-free.

SECTION 03 • SEASONAL FORMULATIONS (4/4)

Winter Season Catalog: Deep Thermal Warmth & Immune Defense

Solar Terms 19 to 24 — Mature red ginger rhizomes, archipelago cloves, and wild forest honey to sustain vitality and peripheral circulation.

No	Solar Term	Blend Formulation	Key Ingredients	Brew Standard	Price	Flavor Profile & Notes
19	 **Start of Winter** 立冬	**Spice Lantern**	cinnamon, clove, dried apple	98°C • 6.0 Min	**Hot 22K** **Iced 24K**	Baked-apple warmth with a little clove.
20	 **Minor Snow** 小雪	**Longan Nest**	longan, red dates, goji	88°C • 4.0 Min	**Hot 28K** **Iced 30K**	Caramel-sweet fruit, no sugar added.
21	 **Major Snow** 大雪	**Ginger Goji**	ginger, goji, lemon	98°C • 6.0 Min	**Hot 18K** **Iced 20K**	The rainy-season favourite.
22	 **Winter Solstice** 冬至	**Solstice Cup**	red dates, ginger, osmanthus	98°C • 6.0 Min	**Hot 24K** **Iced 26K**	Deep, warm and gently floral.
23	 **Minor Cold** 小寒	**Red Date Warmth**	red dates, longan, ginger	98°C • 6.0 Min	**Hot 26K** **Iced 28K**	Rich and mellow.
24	 **Major Cold** 大寒	**Cinnamon Ember**	cinnamon, ginger, orange peel	98°C • 6.0 Min	**Hot 18K** **Iced 20K**	Spicy-sweet, the last blend before spring.

Serving options: Ceramic hot infusion pot (standard) or craft iced shaken tea (+IDR 2,000). Zero synthetic essences; 100% caffeine-free.

SECTION 04 • GUEST JOURNEY & RETAIL FLOW

End-to-End Guest Journey & Atelier Experience

Frictionless 5-phase journey from table seating to savoring handcrafted ceramic tea infusions.

PHASE 1
Table Scan
Instant web menu via camera; zero app install

→

PHASE 2
Customize
Choose from 24 blends, calibrate sweetness & temp

→

PHASE 3
Dynamic QRIS
Amount locked; instant server-side verification

→

PHASE 4
Live Tracking
Transparent real-time brew progression on phone

→

PHASE 5
Ceramic Service
Savor hot artisanal pot infusion at leisure

Experience Dimension	Conventional Cafe Model	CHANGKEER Atelier Standard
Queue Wait Duration	Guests stand in line for 5-10 minutes during peak trading hours.	Zero counter queuing; guests seat immediately and order from their smartphone.
Herbal Ingredient Transparency	Cashiers rarely possess in-depth botanical or functional knowledge.	Interactive digital cards display flavor notes, botanical parts, and health benefits.
Brewing Precision & Consistency	Frequently over-extracted (bitter) when baristas are distracted.	Standardized via digital countdown timers and precise thermal kettle displays.

SECTION 05 • FINANCIAL INTEGRITY & SECURITY ARCHITECTURE

Transaction Pipeline & Server-Side Anti-Tampering

Ensuring banking settlement validity while eliminating client-side price tampering vectors.

STAGE 1
Lock Bill Total
Server calculates total against master DB

→

STAGE 2
Generate QRIS
Dynamic code embeds exact rupiah settlement

→

STAGE 3
Webhook Settlement
Bank gateway issues signed settlement webhook

→

STAGE 4
Dispatch KDS Ticket
Verified ticket pushed to barista queue

Security Safeguard	Engineering Mechanism	Enterprise Operational Protection
Server-Side Anti-Tampering	Pricing is calculated strictly in backend memory; client values ignored.	Prevents fraudulent price overrides via browser inspect tools or modified proxies.
Idempotency Key Protocol	Unique transaction tokens guarantee single settlement execution.	Eliminates duplicate debit charges when guests tap pay repeatedly on lagging connections.
Cryptographic Webhook Signatures	HMAC-SHA256 signature verification on every inbound settlement notice.	Instantly rejects forged payment confirmations or third-party mutation simulations.

SECTION 06 • BAR OPERATIONS & CRAFTSMANSHIP

Barista Standard Operating Procedures & KDS Workflow

Digital queue monitoring ensuring extraction precision, strict timing control, and sweetness balance.

3-State KDS Queue Progression

- **Order Received (Amber):** New incoming order from table; barista checks table number, blend formula, and sweetness level.
- **Brewing Now (Blue):** Barista taps 'Start Brew'; digital extraction timer initiates automated countdown.
- **Ready to Serve (Green):** Infusion completed and poured into pre-warmed ceramic pot; ready for table service.

Thermal Extraction Standard Operating Matrix

Botanical Category	Water Temp	Duration	Quality Standard
Fresh Petals & Green Tea	**80°C – 85°C**	3.5 – 4 Min	Avoids bitter tannins.
Whole Dried Blossoms	**90°C – 94°C**	5 Min	Releases natural color pigments.
Ginger & Turmeric Roots	**98°C – 100°C**	6 Min	Breaks cell walls for essential oils.

Flavor Precision: Natural sweeteners are dispensed strictly using calibrated stainless steel jiggers (0ml, 7.5ml, 12.5ml, 18ml, 25ml).

SECTION 07 • ENGINEERING STACK & RESILIENCE

Technology Ecosystem & High-Availability Architecture

Engineered with modern web standards for high concurrency, banking data safety, and zero downtime.

System Tier	Technology Selection	Architectural Role & Operational Benefit
Frontend & Server Web	Next.js 16 (App Router, React 19)	High-speed server rendering supporting instant guest access across mobile devices without app installation friction.
Programming Language	TypeScript 5 (Strict Mode)	Full end-to-end type safety eliminating runtime exceptions in transactional calculations and recipe parameters.
Interface Design	Tailwind CSS	High-contrast, ergonomic touch interface complying with modern accessibility and editorial standards.
Relational Database	PostgreSQL (via Supabase)	Banking-grade ACID transactional consistency fortified with Row-Level Security (RLS) for data protection.
Data Validation	Zod Schema Validation	Strict schema enforcement across all incoming HTTP order payloads prior to executing server business logic.

Dual-Tier Resilience: Active kitchen orders are cached locally in bar memory, ensuring uninterrupted service and receipt printing during network downtime.

SECTION 08 • BOARD OF FOUNDERS & STRATEGIC VISION

Board of Founders & Operational Governance

Executive synergy and operational dedication behind a modern, transparent herbal tea atelier.

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Project Synthesis: CHANGKEER ATELIER establishes that indigenous botanical wisdom and the 24 Solar Terms can be successfully elevated into a standardized, digitally secure, and commercially viable modern retail model.

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